Ori Livson

Current Roles

Children's Medical Research Institute (ProCan) - *Lead Software Engineer*

Dec 2021 - Present

At ProCan, a cancer proteomics lab, my role centers around the development of their big-data capability, automation of data analysis pipelines, visualisation, user-interfaces, LLM-based applications, specialised HPC environments and CI/CD pipelines.

Additionally, I routinely work with Cancer Data Scientists, Proteomicists and Oncologists to support exploratory data analysis, visualisation and quality control. This is primarily done in my capacity as the lead developer of their project POCKet (Proteomics Oncology Knowledge Base) - a federated query engine and event-driven data warehousing framework.

To enable the above, I have taken a keen interest in stakeholder engagement, communications, training, requirements gathering, and mentorship regarding applicable software development practices across teams.

Key technical skills: Python, Data Science frameworks, Postgres, Trino (FKA Presto), Data Lake Development, Neo4j, Docker & Kubernetes, Vue JS, Graph Databases (Neo4).

University of Sydney - *Associate Lecturer PhD Teaching Fellow*

May 2025 - Present

In this role I support the coordination, development and delivery of lectures and assessments for undergraduate & postgraduate units in the School Computer Science. To date, this includes the units COMP2022: "Models of Computation" and COMP4347/5347: "Web Application Development".

University of Sydney - *Doctor of Philosophy (PhD) - Computer Science*

Mar 2024 - Present

In the Center for Complex Systems, my PhD project "Self-Reference and Incomputability in Collective Decision-Making" investigates novel applications of methods from computability theory to the study of limitative results in social decision making and complex systems theory. For this work, I have been awarded the RTP Stipend Scholarship from 2024-present on a competitive basis.

Prior Employment

Isentia - *Software Engineer*

2020 - 2021 (1.5 years)

At Isentia, a news-media monitoring company, my role centred around software development for natural language processing of voice-to-text transcripts and multimedia ingestion. The highlights of this role are the development of:

- A novel mathematical similarity measure for transcription segmentation jobs. This measure was used to score ML models against each other and human equivalents.

  For demonstrating a model's 15% improvement over its predecessor in closeness to human equivalents, I was a recipient of the 2021 Jeffress Award and Q4 2021 Innovation awards.

- A "capitalisation service", which remedied approximately 90% of incorrect or missing capitalizations by Google voice-to-text transcripts.

- A transpiler from a deprecated text searching language ISYS to ElasticSearch queries.

The latter involved developing AWS serverless architectures to record digital media streams, as well as hardware and software solutions for terrestrial TV and Radio capture across 70 sites in Australia, NZ and Singapore.

Key technical skills: Python, AWS Serverless development esp. API Gateway, ECS, S3, SQS, SNS; Linux System Service Development, NLP.

ezyCollect - *Data Scientist*

2018 - 2020 (2 years)

At ezyCollect, a company specialised in accounts receivables automation, online payments and risk monitoring, I was their Data Scientist hire. My role centred around exploratory data analysis for Marketing and Sales, as well as organising teams to develop and support data products such as insights dashboards and debtor risk profiling.

I also performed all my own software development for Data Warehousing, ETL, Web Development etc.

Key technical skills: Python, Data Science inc. Deep Learning, Clustering, Statistical Tests, Data Engineering, AWS Serverless development esp: AWS Lambda, Glue, Athena, SQS, S3, SQS, SNS; Web Development.

Macquarie University - *Demonstrator*

2017 - 2018 (1.5 years)

Demonstrator for classes over six different subjects on Programming and Web Development for Master of IT Students, Calculus and Linear Algebra for Undergraduates, and a General Interest Mathematics subject about the history and applications of Mathematics in society.

WIN Solutions - *Application Developer*

2015 - 2017 (2 years)

At WIN Solutions, a company specialising in warehouse management software, my role was to develop and support an Android application and hardware solution for labour management, and desktop application development.

Education

Macquarie University - *Master of Research (MRes) - Mathematics*

2016 - 2018 (2.5 years)

Awarded the RTPMRES Scholarship at the 2017 APA rate on a competitive basis to complete a 20000 word thesis on Applied Category Theory under the supervision of Professor Michael Johnson.

My thesis "Least Change View-Updating for Functorial View-Gets with Left Adjoints" explored Category Theoretic formulations of databases, null data and conditions for updatable views with desirable properties.

I was also awarded multiple stipends during the preliminary coursework year (2016) on a competitive basis.

University of Sydney - *Bachelor of Science (Bsc) - Mathematics and Computer Science*

2012 - 2015 (3.5 years)

Mathematics and Computer Science Majors including senior units on AI, operating systems internals, programming language theory, graphics and multimedia. Notable projects include:

- A motion detection program that controls animation. This project was chosen to be shown to the next year's intake of students.

- A Haskell compiler for a simple programming language.

Teaching

COMP5437/4347: Web Application Development
University of Sydney, Semester 1 (2026)

Developed and delivered 6 lectures on web development (under the supervision of Prof Masa Takatsuka).

Topological Social Choice Theory
Chennai Mathematical Institute, Jan-Feb Session (2026)

Developed and Delivered the unit's module on (Combinatorial) Social Choice Theory & Motivations for Topological Social Choice Theory.

COMP2022/2922: Models of Computation
University of Sydney, Semester 2 (2025)

Developed and delivered the unit's Turing Machines Module (under the supervision of Dr Sasha Rubin).

Publications

- Livson, O., Pritam, S., & Prokopenko, M. (2026). The Non-Orientable Topology of Condorcet's Paradox. Mathematics, 14(12), 2127. https://doi.org/10.3390/math14122127

- Prokopenko, M., Ay, N., Breviario, A., Crocker, R.M., Davies, P.C., Davies, P.C., Dougan, D., Fletcher, R., Harr'e, M., Heisler, M.G., Kuncic, Z., Lewis, G..., Livson, O., Reiner, V., & Serra, J. (2026). Algorithmic bottlenecks in evolution: Genetic code, symbolic language, and the Great Filter hypothesis. (arXiv)

- Nawaz, U., Deng, N., Livson, O., Mayoh, C., Lau, L. M. S., Reddel, R. R., ... Poulos, R. C. (2026). OnCorr: A pan-cancer mRNA-protein correlation tool for precision oncology. Npj Precision Oncology, 10(1). doi:10.1038/s41698-026-01323-2

- Prokopenko, M., Davies, P., Harré, M., Heisler, M., Kuncic, Z., Lewis, G., Livson, O., Lizier, J., & Rosas, F. (2025). Biological arrow of time: emergence of tangled information hierarchies and self-modelling dynamics. Journal of Physics: Complexity, 6(1), 015006.

- Livson, O., Prokopenko, M. (2025). Arrow's Impossibility Theorem as a Generalisation of Condorcet's Paradox (arXiv)

- Livson, O., Prokopenko, M. (2025). Comparing and Contrasting Arrow's Impossibility Theorem and Gödel's Incompleteness Theorem (arXiv)

Additional Projects and Interests

- Development of my personal website https://ori-livson.com, where I write tutorials on software engineering and mathematics.
- Stock Fundamentals Analysis using Serverless Web Scraping (active since 2019)
- Keen interest in functional programming, esp. Haskell and Elm.
- Arduino development and basic electronics.
- 3D Modelling with Blender